HK IMO Training 2021-2022
Problem Set 3
For 9 October 2021

Problem 9. Let AB be a diameter of a circle Γ. Another circle with centre B intersects Γ at C and D, and intersects the diameter AB at P. A line passing through P meets Γ at E and F such that D, F, A, E, C lie in this order on Γ. The line CP intersects the segment AF at X, and intersects the ray AE at Y. Prove that A, X, D, Y are concyclic.

Problem 10. Let n be a positive integer. In a country, there is a king and 2^n citizens. Every citizen has an infinite number of banknotes and a finite number of coins. The value of a banknote is \$$2^n$, and the value of a coin can be one of \$1, \$2, \$$2^2$, ..., \$$2^{n-1}$. Let S be the total number of coins in the country. The following policy is implemented.

- Every day, each citizen must give a certain amount of money to either another citizen or the king using the banknotes and coins that the citizen has.
- The amount of money given by each citizen is \$1 more than the total amount of money he/she receives on that day.
- The citizens can only give away money that they currently have. Money received on that day cannot be used immediately. Also, no change is allowed.

Find the smallest possible value of S such that this policy can be implemented indefinitely (given that all citizens are cooperative).

Problem 11. Find all function(s) $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x + f(y)) = yf(xy + 1)$ for all $x, y \in \mathbb{R}^+$.

Problem 12. Let n be a positive integer. Suppose p is a prime divisor of $n^3 - 3n + 1$. Prove that $p = 3$ or $p \equiv \pm 1 \pmod 9$.